

Table 2. **Operational definitions used to place each work.** The stages (S0 to S4, plus the consumers X; theory and analysis works are listed but sit outside the core) are defined by *where the calibration signal enters the decision pipeline*, so a work is placed by what its main contribution changes, not by the task it is evaluated on. The first row defines a term, not an inclusion rule: works that report only discrimination are included and marked by their metric. The last three rows define properties recorded for every work: the Contract, Action and Guar. columns of Tables 3 and C1, where Action records what the rule or trained behaviour does when the confidence is low.

Construct	Counts if and only if...	Does not count
Calibrated confidence	the confidence is scored against empirical accuracy with a calibration metric (ECE, Brier, NLL or a reliability diagram)	a confidence that is only reported, or only used to rank answers (AUROC measures discrimination, not calibration)
S0 read-off	the confidence is the model's own token or sequence probability, unchanged	any map applied on top, including a probe on the hidden states (S1)
S1 post-hoc map	a map fitted on held-out data [Guo et al. 2017], on top of the read-off, as a probe on the hidden states of the frozen model, or as an auxiliary model reading the input and output text, or a fixed map (such as pooling the probability of synonymous labels or reweighting token probabilities), changes the confidence but not the model's parameters	re-training or fine-tuning the model (S4)
S2 prompt elicitation	the model is asked, in its input, to state a confidence in words or numbers [Tian et al. 2023]; a method whose main contribution is the prompt stays here even if it also aggregates several prompted answers	a confidence computed outside the model's own text (S0, S3)
S3 extra inference	the confidence needs extra inference, repeated sampling or perturbation (including internal-state signals compared across samples), or carries a distribution-free calibration: a prediction set or a threshold calibrated by conformal or risk control [Campos et al. 2024; Farquhar et al. 2024]	a single forward pass, read off (S0) or probed (S1)
S4 training	a calibration term appears in the model's own loss or reward: a confidence target or calibration loss, a calibrated reward model, a calibration reward, an abstention target, or a step- or meaning-level confidence reward (families T1 to T5) [Bani-Harouni et al. 2026b; Leng et al. 2025]	calibration that changes only as a side effect of training for something else; theory and analyses of such training, which are listed separately
X consumers and contract	the work uses a confidence to act (abstain, hand off, retrieve, filter, adapt compute, set), fixes the output contract, calibrates a model used as a judge against human labels, studies how people rely on a model's output or its stated confidence, or evaluates or benchmarks confidence-based abstention, without changing how the confidence is produced	a work whose main contribution is the confidence itself (S0 to S4)
Theory and analysis	the work analyses when or why training changes calibration, without proposing a training objective	a training method, even one motivated by such analysis (S4)
Typed decision output	the options are declared before inference and the model assigns a probability to each option, so the decision and its confidence come from the same distribution	a single output generated under a grammar or schema (constrained output); free text parsed after the fact

Table 2, continued

Construct	Counts if and only if...	Does not count
Action policy	an explicit rule, or a behaviour the model is trained into (such as refusing when unsure), maps the confidence to an action: answer when it is high; when it is low, abstain, hand the case off to another model or a human (which the papers call routing, cascading, deferral or escalation), trigger retrieval, filter the output (remove, rewrite or revise low-confidence claims, or keep the most confident of several sampled answers), adapt compute (keep sampling, reasoning, querying or exploring, or spend a larger budget; the same rule stops early when the confidence is high), or return a prediction set	a confidence reported with the answer or displayed to a person, with no rule or trained behaviour attached, or a selective-prediction curve reported only as an evaluation (recorded as none; the Metric column records the metrics of the main text)
Coverage guarantee	the method states a finite-sample, distribution-free bound, such as conformal coverage of at least $1 - \alpha$ [Campos et al. 2024]	good empirical calibration on a benchmark, without a stated bound